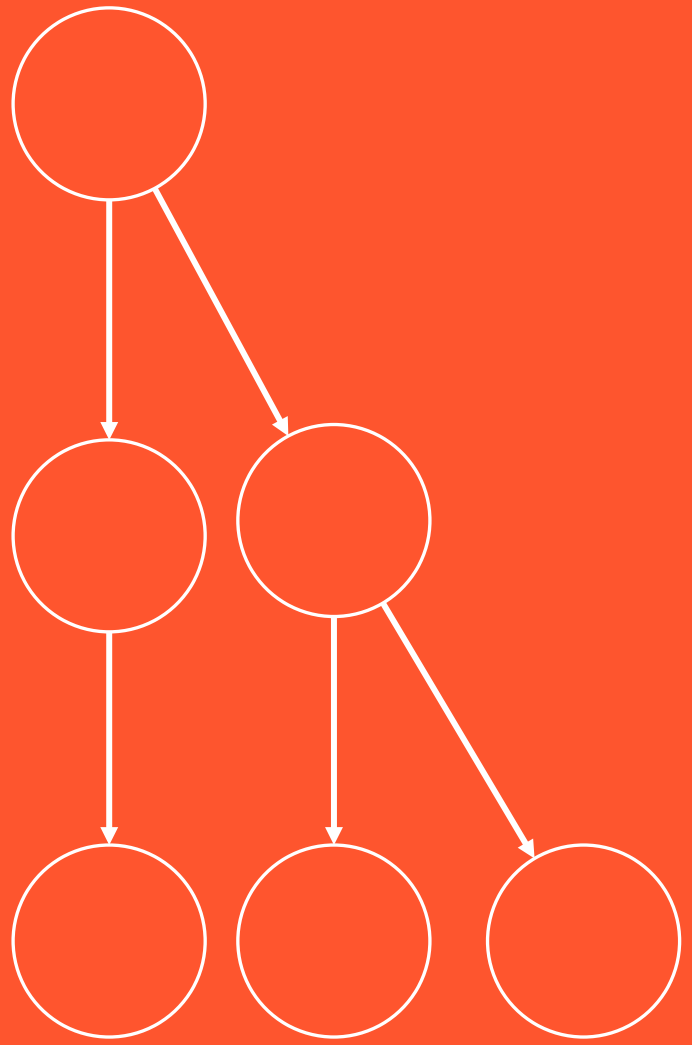


BCOG 100

Social Cognition

Lecture 11.1 — Living in Lasting Groups

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Memory and Social Life

In your group, three minutes.

Last week, we talked about memory and generative models.

This week, we are talking about life in complex, hierarchically organized social groups. Why might complex memory a prerequisite, or at the very least a huge facilitator, for the evolution of complex social groups?

A Morning in a Baboon Group

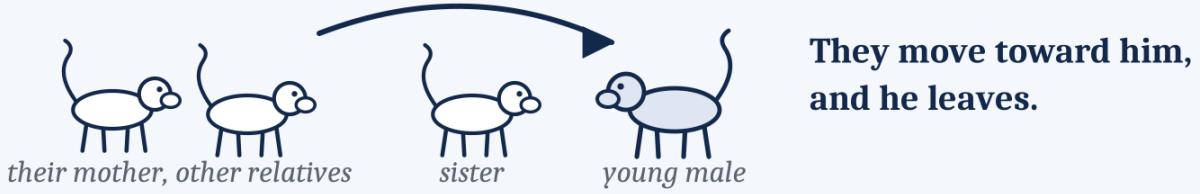
Almost everything a baboon does has an effect that depends on what the other baboons do.

A morning in a baboon group

1. A higher-ranking female walks toward the fig tree.



2. Later, a young male threatens her sister.



What a baboon does next depends on

what, and who, it has seen

what it is trying to get

what has happened between it and the others before

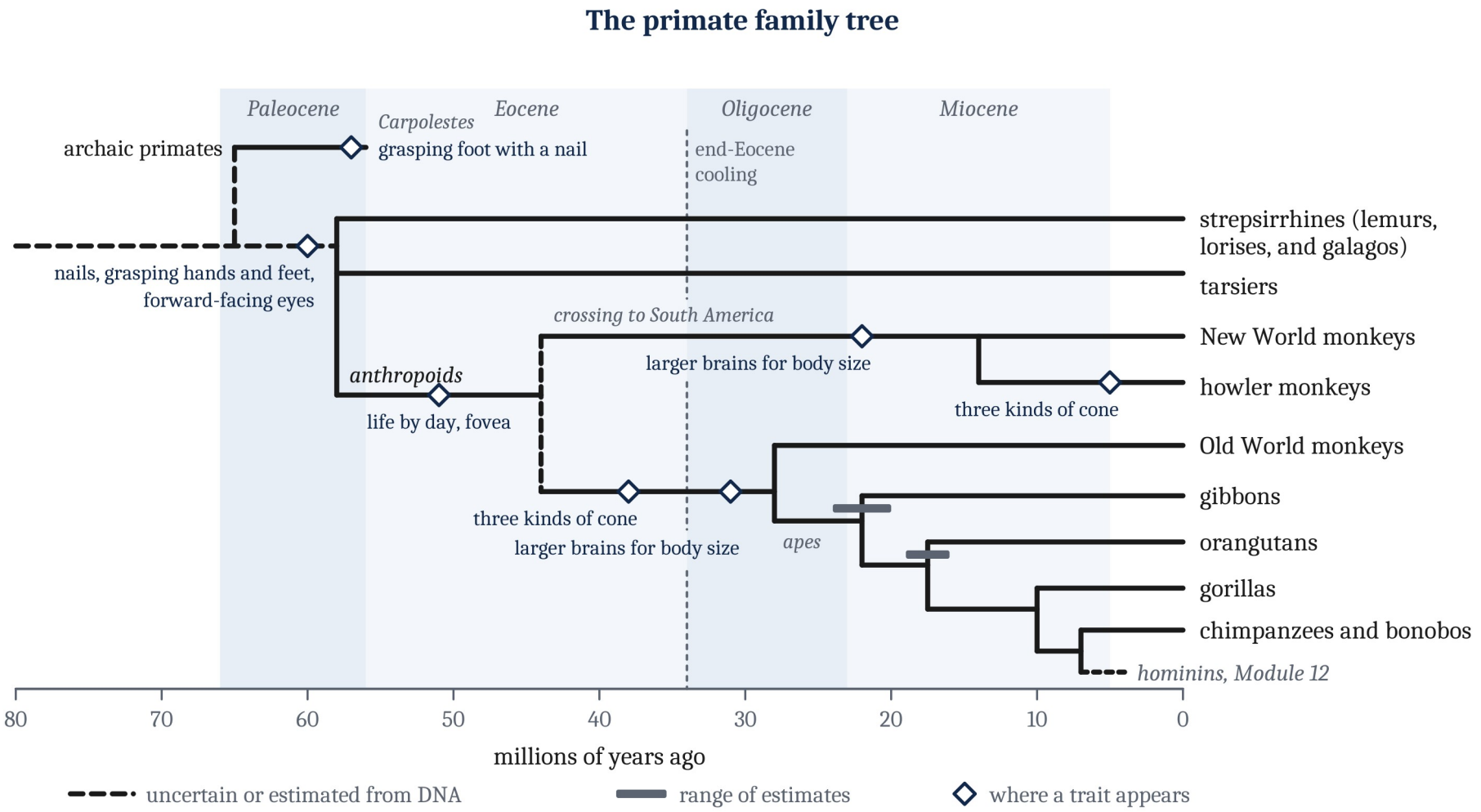
None of these can be observed directly.

Her most important surroundings are the other members of her group.

An illustration of a typical morning, from the reading's account of baboons in southern Kenya

A New Kind of Animal

Anthropoids: the monkeys and apes, including human beings. They are one of three primate lines, and all three have evolved for equally long.

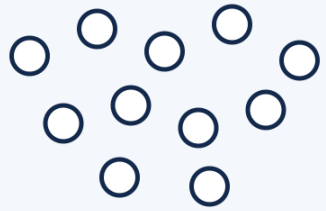


Lasting Groups and Larger Brains

Most primates live in permanent groups of both sexes. For their body size, monkeys' and apes' brains are about twice as large as most mammals'.

Lasting groups

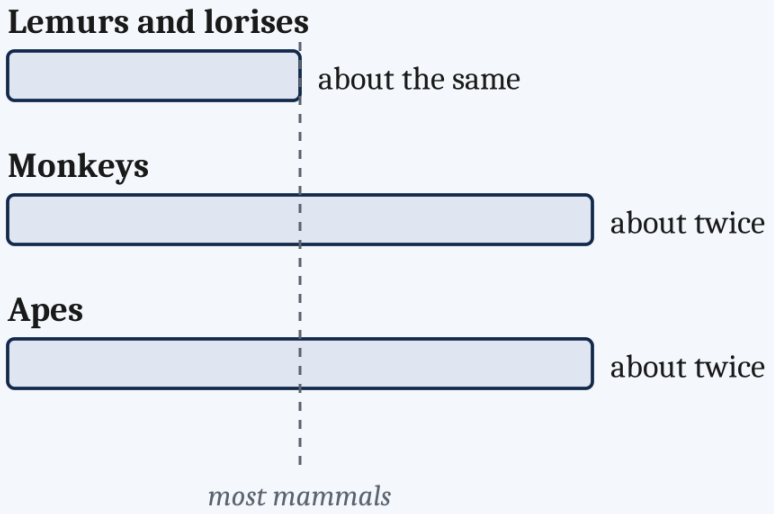
The great majority of primates live in permanent groups of both sexes, with three or more adults. This is unusual among mammals.



Living alone is not the same as being unsocial: primates that forage alone still have social relationships.

Brain size for body size

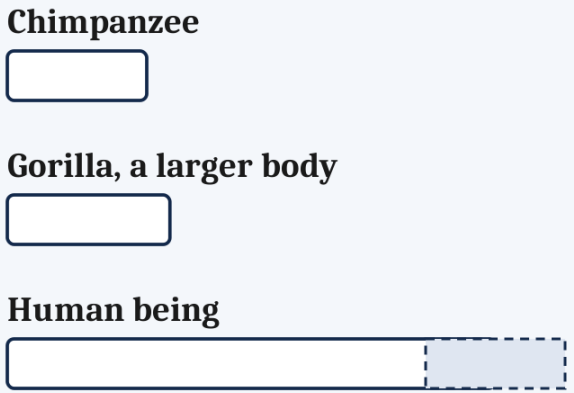
relative to most mammals



Body size accounts for about three quarters of brain-size differences.

Human beings are the exception

absolute brain size



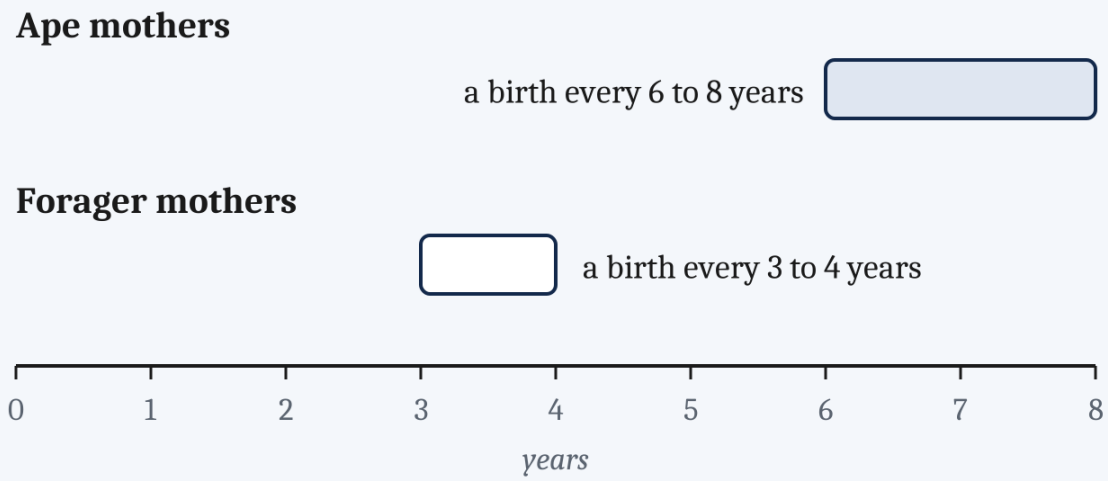
Human: three to four times a chimpanzee's.
Gorilla: about a third of a human's.

Lasting groups also arose separately in dolphins, elephants, hyenas, and crows: convergent evolution

A Slow Life and a Long Dependence

Life history: the timing of growth, maturity, reproduction, and death across a lifetime. An **alloparent** is an individual other than a parent that cares for the young.

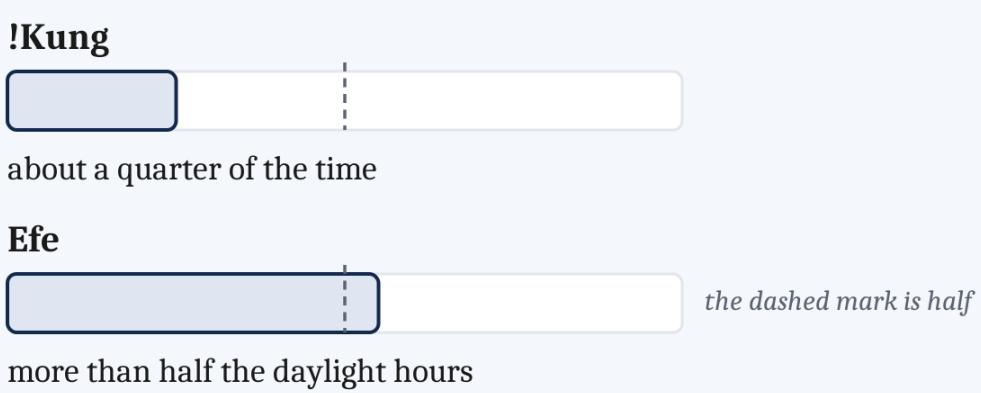
Time between births



Raising a child until it feeds itself: on the order of 13 million calories.

Hrdy argues that births this frequent are possible only because others help.

Others hold the infant



An alloparent: an individual other than a parent that cares for the young.

A long dependence goes with a large brain: for about a year after birth, the human brain keeps growing at the rate at which it grew before birth.

Caring for the Young

In **cooperative breeding**, individuals other than the parents regularly help care for and feed the young. Both sexes choose mates and compete for them.

Who cares for the young

Mammals: mothers give most care;
no mammal has care by fathers alone.

Birds: usually both parents.

Fish that give care: most often
the father alone.

Among primates

About half of species: others hold
and carry infants.
About a fifth: others also give food.
Titi monkeys: fathers carry infants
for most of the day.

Cooperative breeders: marmosets,
tamarins, and human beings.

Both sexes choose and compete

Which sex is more selective, and
which competes more, varies.
66 species: more mates went with a
greater increase in offspring for males
than for females, on average; most
with care by females alone;
no difference detected with care
by males alone.
Infanticide: by newly joined males,
and by females, probably more
often among mammals.

Which sex is heavier, across mammal species



In most lemurs, lorises, and galagos, the sexes are similar in size.

Helping Relatives

Relatedness: the probability that two individuals share a gene through descent from a common ancestor. **Kin selection**: selection on a trait through its effects on relatives that share the trait.

The puzzle

Selection favors genes that lead to more copies of themselves.

Help at a cost to the helper seems to do the opposite.

Yet animals in groups groom one another, share food, call out at a predator, and join one another's fights.

Relatedness, r

full sibling **r = 1/2**

help is favored only if the gain is more than 2 x the loss

half-sibling **r = 1/4**

help is favored only if the gain is more than 4 x the loss

first cousin **r = 1/8**

help is favored only if the gain is more than 8 x the loss

Kin selection: r b > c

benefit to the relative, times r, exceeds the cost to the helper

Kinship is uneven

average relatedness among resident females

Chimpanzees

below 0.01

Yellow baboons

0.09

Meerkats

0.42

Relatedness cannot be observed directly.

Hamilton: in a world of model animals, none would give its life for one relative, but each would for "more than two brothers, or four half-brothers, or eight first cousins"

Kin selection explains some of the help in primate groups, but not all of it

The Prisoner’s Dilemma

A **payoff matrix** gives what each of two individuals gains or loses for each combination of what they do; the study of such situations is **game theory**. In the **prisoner’s dilemma**, each does best by defecting while the other cooperates, yet both do worse if both defect than if both cooperate.

A table from Module 7

A small fish deciding whether an eel is present: four outcomes of one decision.
hit, miss,
false alarm,
correct rejection

Here, the outcome depends on what two animals do.

Player A \ Player B	cooperates	defects
cooperates	**3, 3**	**0, 5**
defects	**5, 0**	**1, 1**

points for A, then for B

Reading the table for A

If B cooperates: A gets 5 by defecting, 3 by cooperating.

If B defects: A gets 1 by defecting, 0 by cooperating.

Each does better by defecting whatever the other does, so both end at 1, 1, when both could have had 3.

Cooperating is good, but taking advantage of the other is better.
This helps explain why selfish behavior can persist.

Hawks, Doves, and a Stable Mix

In game theory, a **strategy** is what an individual does in each situation that a game can present, whatever produces that behavior. An **evolutionarily stable strategy**, once common, cannot be bettered by any rare alternative.

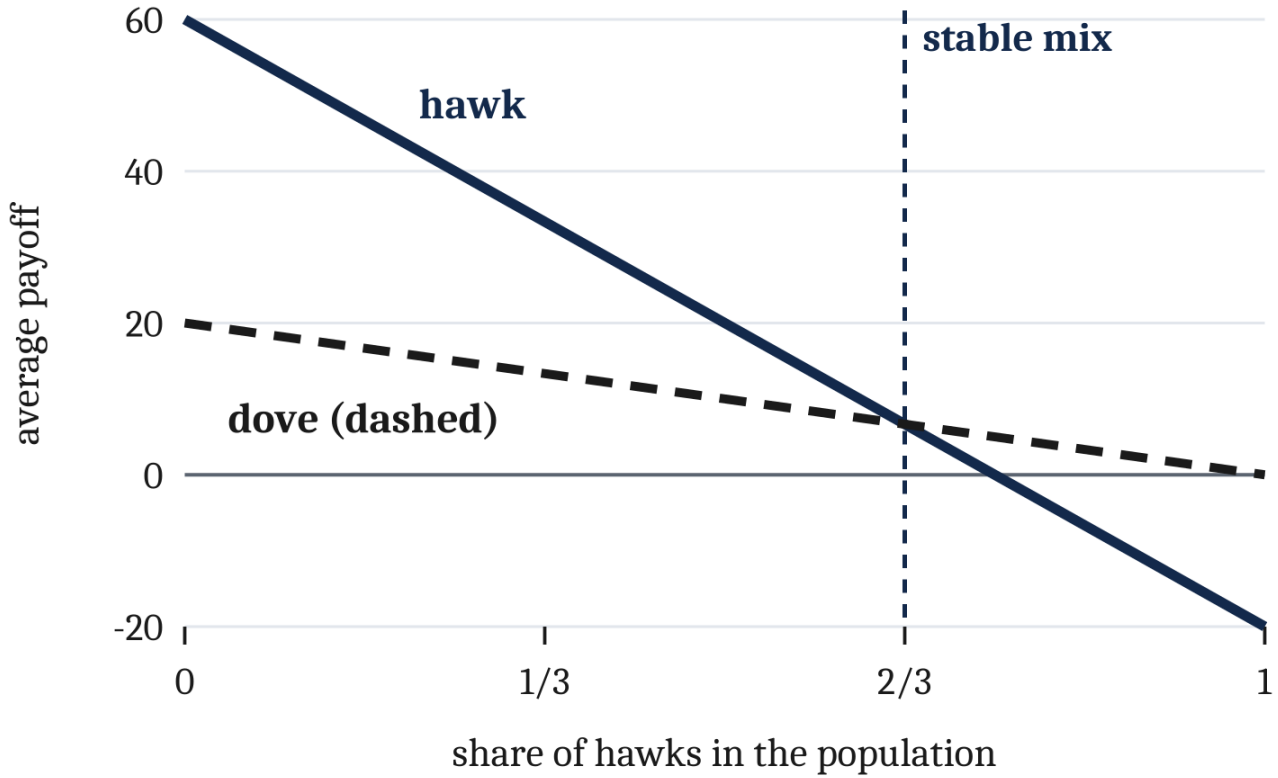
The hawk-dove game

payoff to the animal named in each row

	vs. a hawk	vs. a dove
hawk	**(V - W) / 2**	**V**
dove	**0**	**V/2 - L**

V = the value of the resource
W = the cost of an injury
L = the cost of a long display

Hawks and doves settle at a stable mix



V = 60, W = 100, L = 10

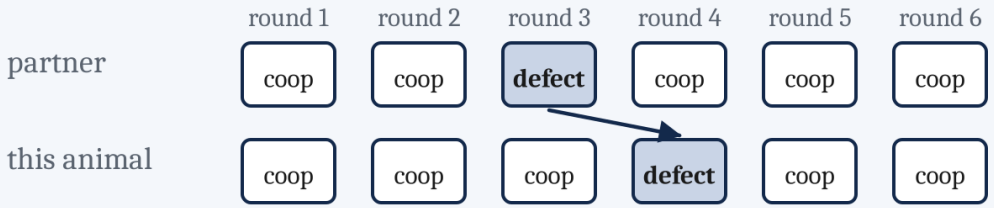
Repeated Encounters

Reciprocal altruism: help given by one individual now and returned by another later, between individuals that need not be related. When the same partners encounter one another again and again, repetition changes the game.

One round, and many rounds

In one round, defecting does best.
In a lasting group, the same partners encounter one another again and again.

An example: cooperate only with a partner that cooperated the last time.



What each gains depends on how each responds to what the other did before.

Vampire bats

In the wild: most blood sharing is mothers feeding their own young; the rest goes mainly to relatives and to roostmates.

In captive groups of wild-caught bats: whether one bat had fed another earlier predicted whom it fed far better than relatedness did.

Still open: exchanges are loosely matched, and no study has tested how bats respond to a partner that fails to feed in return.

Evidence for reciprocal altruism outside human beings is thinner than is often claimed.

Life in a Primate Group

A **dominance hierarchy** is a stable ordering of who gives way to whom in contests.

Four parts of a social system

Social organization

How large is the group, and who is in it?

Social structure

What relationships do members have?

Mating system

Who mates with whom?

Care system

Who cares for the young?

after Kappeler (2019)

Baboons

Females stay for life; males leave when they mature.

Females form a dominance hierarchy; a daughter ranks just below her mother.

Strongest bonds: mothers and daughters, and sisters.

Chimpanzees

A community of about 20 to 150 splits into parties that join and separate.

Males stay; females usually leave. Males form alliances and rank in a hierarchy.

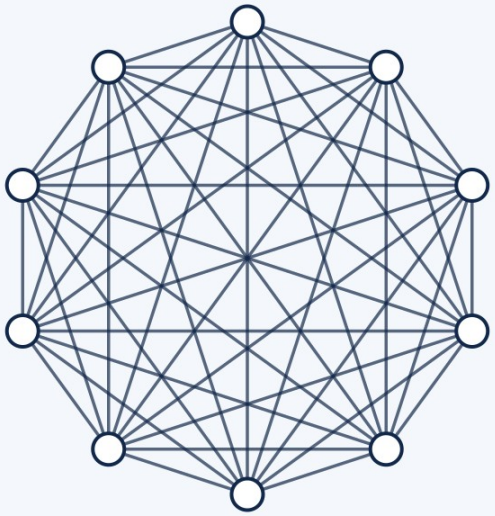
At seven field sites, 39 behaviors were common in some communities and absent in others, not explained by habitat.

Competing for food, a subordinate chimpanzee goes for food hidden from the dominant's view, and leaves alone food that the dominant has a clear view of.

Why Predicting Others Is Hard

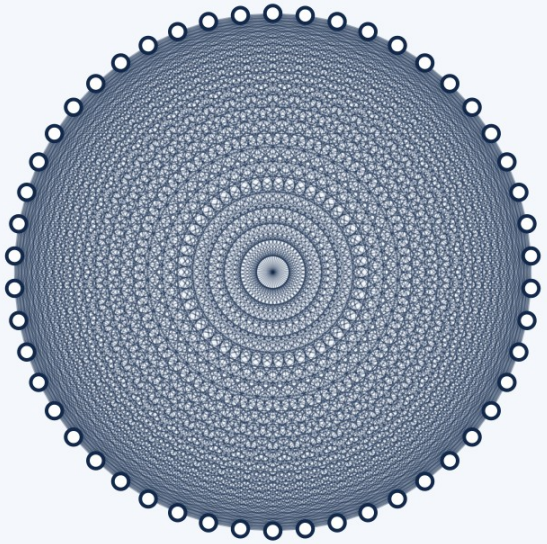
What another animal does next depends on what cannot be observed directly. And the number of pairs in a group grows roughly with the square of its size.

10 animals



45 pairs

50 animals



1,225 pairs

What can be observed

Another animal's behavior, where it is looking, and what happened on earlier occasions.

What cannot be observed directly

What it has seen, what it is trying to get, and what has happened between the two of them before.

Which family it belongs to predicts some of what it will do.

n animals: each deals with n - 1 others, but there are n(n - 1)/2 pairs, and the pairs affect one another.

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